

What is inside OLED

In an OLED, the emissive electroluminescent layer is a film of an organic compound that emits light in response to electric current. Small molecules, oligomers, polymers and organometallic species and other variety of compounds have been used for OLEDs. High-efficiency emission processes are the key requirements for emissive materials.

Organometallic chelates, fluorescent and phosphorescent dyes and conjugated dendrimers can be used to make small molecules. To obtain light emissions at different wavelengths, fluorescent dyes can be used. Compounds such as perylene, rubrene and quinaclidone derivatives are often used. Derivatives of poly (p-phenylene vinylene) and polyfluorene are the typical polymers used in polymer LED (PLED) displays. Due to their fast response time, low voltage, high luminous efficiency and viewing-angle performance, these are used more in flexible and free-shape devices.



Aesthetics with lighting

Successful reduction in the candidate pool. Machine-learning tools have helped predict the colour and brightness of molecules using simple quantum chemical calculations. Finally, these were left with hundreds of blue molecules with performance improvement over organometallic molecules.

Harvard's Office of Technology Development has a pending patent application on the molecules while they consider commercialisation opportunities. Further research in the field is expected to lead to molecules that could be used in flow batteries, solar cells, organic lasers and more.

Thompson says, "We are now looking at an approach that can capture high energy before the molecules disintegrate." If we could stabilise reactive states, it could open up a lot of possibilities. Let us find out what is degrading the blue molecules, and what could possibly lead to some very interesting displays in the future.

TADF could hold another answer. According to experts, thermal-

ly-activated delayed fluorescence (TADF) could solve the challenge of efficient and stable blue emission. Initial TADF devices, however, lost five per cent of their brightness in just 85 hours, which fuelled further research into the process.

Claims have been made regarding extending this time by eight times since then. In the newly-developed modification, two extremely thin (1nm to 3nm) layers of lithium-containing molecule, Liq, are placed on each side of the hole-blocking layer. This brings electrons to TADF material and prevents holes from leaving before contributing to emission.

A team led by Prof. Tae-Woo Lee, department of materials science and engineering, Pohang University of Science and Technology, South Korea, used pure organic TADF emitters that show very high internal quantum efficiency of nearly 100 per cent without precious metals. Easy synthesis due to use of pure organic molecules reduces synthesis cost. This is a big leap towards the development of inexpensive and solution-processed OLED displays and solid-state lighting.

Pinning together different refractive indices of titanium oxide and a conductive polymer, composite electrodes were developed by a collaboration of Korea Advanced Institute of Science and Technology, South Korea, and Pohang University of Science and Technology researchers. This increased the optical efficiency to about 1.5 times.

Usually, increasing the efficiency leads to a drop in flexibility. However, due to the involvement of a composite electrode, it endured strains four times stronger than convention. Even after 1000 bending cycles at a radius of curvature as small as 2.3mm, OLEDs remained intact and operational.

Achieving transparency and flexibility

Wearables and curved displays are seeing a lot of demand in the current market, creating a requirement for flexible displays. Graphene based OLEDs have reported the best efficiency and almost the same level as indium-tin-oxide based OLEDs.

Graphene as a highly-flexible light source promises to be a key element in next-generation displays and lighting. Current flexible solutions involve graphene based OLEDs, which use graphene as a transparent electrode placed between titanium-dioxide and conducting polymer layers. Efficiency and effective reflectance of transparent electrodes is increased in optical designs, which is due to the synergistic collaboration between high-n and low-n layers.

If all goes well, we could very well see foldable smartphones from Samsung employing flexible OLEDs as early as next year. Lenovo showed off a phone and tablet at its tech world convention in San Francisco, USA, this year. Youtubers prototyping the device can be seen wrapping these around their wrists.

Involvement of ceramics has been a more recent development. Dutch R&D institute, Holst Centre, in collaboration with American ultra-thin ceramics supplier, ENRG Inc., has designed and developed a flexible OLED that promises a ten-year-plus lifetime. The 12cm x 2.5cm prototype offers an intrinsic barrier to protect OLEDs from the environment.

Most flexible OLEDs require a